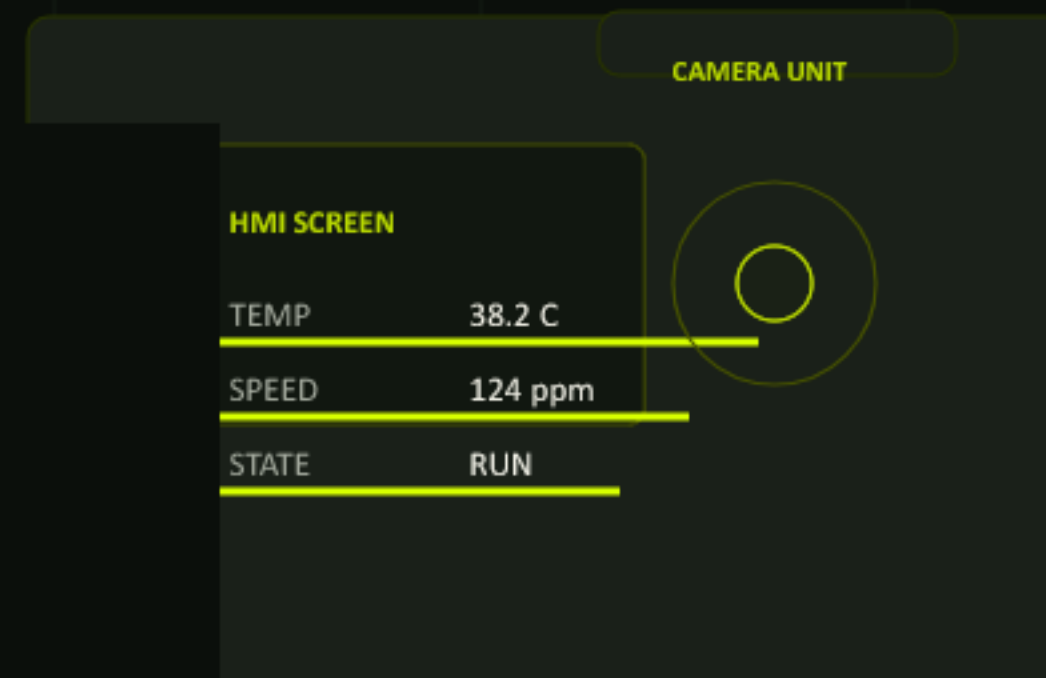


// HMI MONITORING PLATFORM

CAMERA-BASED HMI VISIBILITY.

PingPong Vision reads production-machine HMI screens with a camera and AI OCR, then turns the readings into live dashboard data.



CAMERA

INPUT

AI OCR

FIELD READING

DB

HISTORY + EVENTS

// PROBLEM

MACHINE DATA IS TRAPPED BEHIND HMI SCREENS.

Older machines and vendor-controlled platforms may not expose PLC/API data. The visible screen is often the only universal interface.

01 No PLC integration

The system avoids control commands, OPC UA tags, and machine automation logic.

02 Works across brands

A camera can observe Siemens, Multivac, legacy panels, and analog-like interfaces.

03 Creates history

Visible values become structured readings, events, and dashboard trends.

// CORE WORKFLOW

ONE CAMERA TURNS SCREENS INTO DATA.

The operator configures the HMI once, then the worker repeatedly reads the configured fields from live frames.

01 Capture HMI

USB camera streams the machine screen locally or through the relay.

02 Calibrate fields

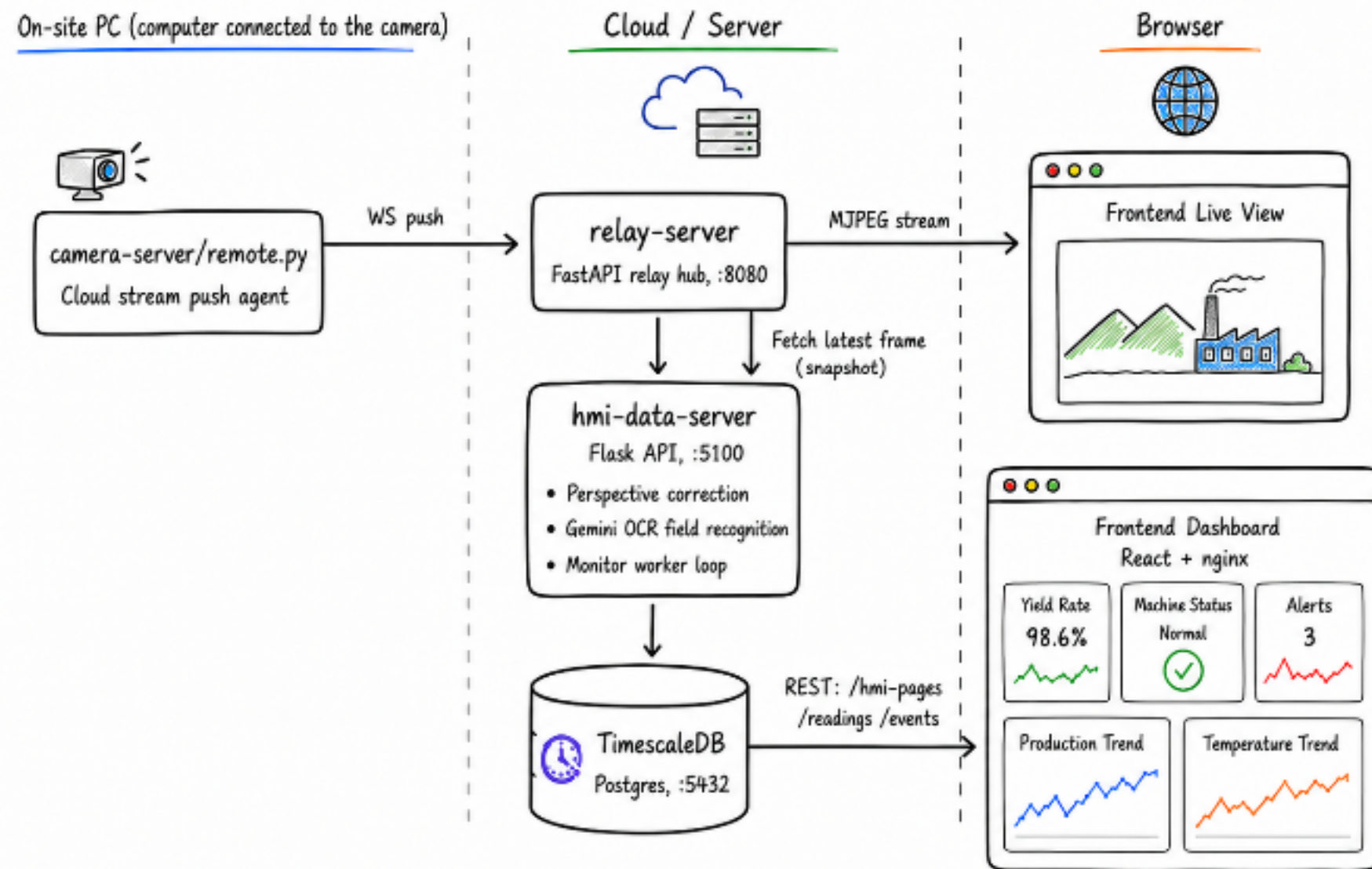
Onboarding stores screen corners and field bounding boxes.

03 Read + persist

The worker applies perspective correction, calls Gemini OCR, and saves readings.

PINGPONG VISION — HMI Monitoring Platform

An industrial HMI remote monitoring system based on cameras and AI OCR



// SERVICE RESPONSIBILITIES

EACH SERVICE OWNS ONE PART OF THE VISION PIPELINE.

camera-server

Runs on the camera PC; exposes LAN MJPEG or pushes frames to relay.

relay-server

Buffers the latest frame per camera and fans out MJPEG to frontend and worker.

hmi-data-server

Business REST API; owns HMI pages, fields, readings, events, monitor jobs.

hmi-monitor-worker

Single process that actually runs OCR loops and writes readings.

TimescaleDB

Stores HMI pages, field definitions, time-series readings, events, worker status.

// DATA MODEL

HMI DATA LIVES IN THE BACKEND.

The browser owns only temporary session state. HMI business data is loaded from and written to the HMI data server.

hmi_pages

Configured camera/HMI page or machine location.

hmi_fields

Detected labels, units, visual types, and bbox_percent.

hmi_readings

Time-series OCR values, confidence, latency, metadata.

hmi_events

Image-change, operational, and field-threshold incidents.

hmi_monitor_jobs

Desired monitor state plus worker heartbeat/status.

MONITORING IS MORE THAN OCR VALUES.

01 Image-change AI

Compares baseline and live frames to detect hands, moved camera, page changes, or value changes.

02 Operational events

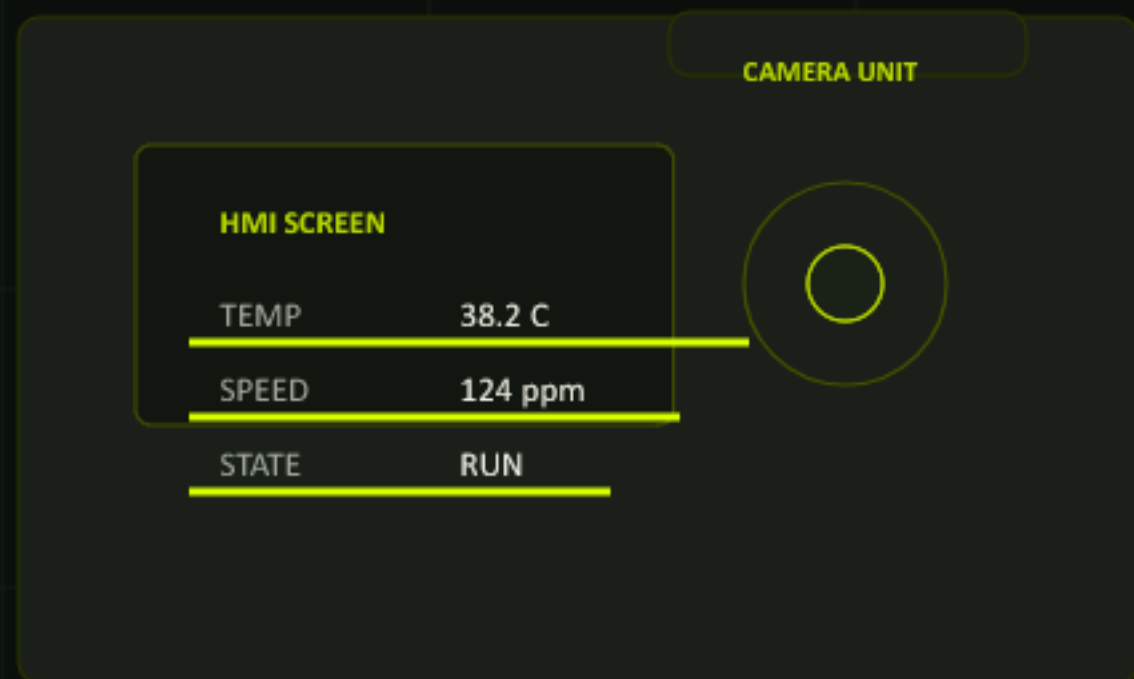
Worker records `camera_connection_lost` and `repeated_unreadable_values`.

03 Field thresholds

Dashboard rules can turn readings into warnings and alerts.

Known constraint: calibration depends on camera alignment and HMI page stability. If the camera moves or the screen layout changes, the stored field boxes can stop matching the live image.

// PROJECT TAKEAWAY



PINGPONG VISION

**READS THE MACHINE SCREEN,
NOT THE MACHINE CONTROLLER.**

The project demonstrates a non-invasive path to production visibility: camera capture, AI OCR, time-series storage, event logging, and a dashboard for HMI-based machine monitoring.